Engineering Project Report

Project 066: Digital Thermometer Circuit Using LM35 Temperature Sensor

Bigyanlabs Engineering Team
Electronics & Embedded Systems Division

August 11, 2026

Abstract

This technical report presents the comprehensive design, circuit schematics, bill of materials, mathematical modeling, assembly methodology, and experimental testing for **Digital Thermometer Circuit Using LM35 Temperature Sensor**. Classified under the **Intermediate (College Level)** tier in the **Precision Temperature** category, this design provides optimized performance, high reliability, and clear practical utility for school and engineering applications.

Contents

1 Introduction & System Overview

The objective of this project is to implement a robust electronic system for **Digital Thermometer Circuit Using LM35 Temperature Sensor**. This document details the step-by-step engineering design, component functions, mathematical principles, and experimental results.

2 Bill of Materials (Components List)

Table 1: Bill of Materials for Digital Thermometer Circuit Using LM35 Temperature Sensor

S.No.	**Component Name**	**Qty**	**Circuit Function**
1	Microcontroller / Op-Amp	1	Arduino UNO / ESP8266 / ESP32 / LM358
2	Transducers / Sensors	2	Interface sensor modules
3	LCD / OLED Display	1	Visual data display (16x2 / I2C)
4	Relays / Driver MOSFETs	2	Actuator control interface
5	Passives and Connectors	1 Set	Resistors, capacitors, breadboard / PCB
6	Power Module	1	5V / 12V regulated DC source

3 System Architecture & Methodology

The Digital Thermometer Circuit Using LM35 Temperature Sensor integrates hardware sensors and microcontrollers/op-amps. Sensor signals are processed through custom firmware or analog conditioning stages and displayed locally or streamed over Wi-Fi/Bluetooth.

4 Circuit Assembly & Testing Procedure

1. Place core active components and microcontrollers/ICs on the breadboard or PCB.
2. Wire DC supply rails (+VCC and GND) with appropriate decoupling capacitors.
3. Connect sensor networks and input signal conditioning circuits.
4. Interface output switches, MOSFET drivers, or visual/acoustic indicators.
5. Apply power and measure operating node voltages to verify proper performance.

5 Mathematical Derivations & Governing Equations

$$V_{\text{ADC}} = \frac{\text{ADC\_Value}}{1023} \cdot 5.0\,\text{V} \tag{1}$$

$$\text{Distance} = \frac{\text{Time} \times 0.0343\,\text{cm/s}}{2} \tag{2}$$

6 Applications & Engineering Conclusion

Engineering laboratory projects, IoT sensor nodes, automated environmental monitoring, and embedded precision temperature.